Econ 605 Notes

Parker Howell

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General

- GDP = flow. "bricks per quarter"
- Y_t = (stock/flow) total output - not always GDP
- C_t = (stock/flow) consumption
- K_t = (stock/flow) capital - sometimes includes IP, sometimes just physical
- X_t = (stock/flow) QUANTITY of capital...
- $m = e^{-\bar{r}(s,t)}$: discount factor - sometimes with integral????

Solow

- $\dot{A}_t$: (flow) wealth??

OLG

- ??

Ramsey

Planner's Problem

- H: total number of households
- l_t: household size (exogenous)
- $L_t = 1 * l_t * H$ (1 = labor supply per person)
- $C_t = Hx_t$
- x_t: household consumption - CONTROL! (state is K) but we replace this with:
- $c_t = \frac{C_t}{e^{gt} L_t}$
- $k_t = \frac{K_t}{e^{gt} L_t}$
- ρ: time discounting (impatience)
- θ: IES intertemporal elasticity of substitution
- $\theta = -\frac{u_c(c)}{c \cdot u_{cc}(c)} = -\frac{d \ln c}{d \ln(u'(c))}$
- $g_Y = n + g$
- $\frac{\dot{c}}{c} + g$: growth rate of per-capita consumption
- $\vec{\alpha} = (\rho, \theta, n, d, g)$ ALL ARE EXOGENOUS
- $PD(\vec{\alpha})$: phase diagram for comparative dynamics

From Optimal Control Slides:

- $\bar{k}$: max debt constraint or min wealth constraint IN TERMINAL STATE
- Γ: law of motion for capital/wealth/the state variable: $\Gamma(k_t, c_t, t) = \dot{k}$
- λ_t: marginal benefit of one unit of k_t
- $\lambda_t k_t$ = agent's wealth measured in units of utility (i.e., potential to receive future utility)

LOM for capital:

$$\dot{K} = I - dK = Y - C - dK$$

LOM for capital, intensive form:

$$\dot{k_t} = y_t - c_t - (n+d)k_t = \underbrace{f(k_t) - c_t}_{\substack{\text{actual investment} \\ \text{per effective worker}}} - \underbrace{(n+d+g)k_t}_{\substack{\text{replacement investment} \\ \text{per effective worker}}}$$

Production Function/Total Output

$$Y = F(K, e^{gt}L) = C_t + I_t$$

Planner's Problem

$$\underbrace{V(k_0, t)}_{\text{indirect utility}} = \max_{c_t} \int_t^\infty e^{\rho t} e^{nt} u(c_t) dt \text{ s to}$$

1. $\dot{k} = y_t - c_t - (n+d)k_t$
2. $k_0 > 0$ given
3. $\lim_{t \to \infty} \lambda_t k_t$ (transversality condition with exponential discounting)
4. implicitly: $k_t, c_t \geq 0$ - will never bind due to Inada so we usually ignore them

Planner's Problem FOCs:

- $\frac{\partial \mathcal{H}}{\partial c_t} = 0$
- $\frac{\partial \mathcal{H}}{\partial k_t} = -\dot{\lambda}$
- $\frac{\partial \mathcal{H}}{\partial \lambda_t} = \dot{k_t}$

Euler Equation for Consumption (Solution to the Ramsey Problem)

$$\frac{\dot{c}_t}{c_t} = \frac{1}{\theta}(r_t - (\rho + \theta g))$$

Phase Diagram Equations (steady state)

$$\dot{k} = 0 \iff c_t = f(k_t) - (n + d + g)k_t$$
$$\dot{c} = 0 \iff k_t = k_*, \text{ where } f'(k_*) = d + \rho + \theta g$$
$$c_* = f(k_*) - (n + d + g)k_*$$

Golden Rule vs Steady State Capital

- $f'(k_*) = d + \rho + \theta g$
- $f'(k_G) = d + n + g$

So

$$f'(k_*) > f'(k_G) \iff \rho > n + (1 - \theta)g$$

Isoclines

- $\dot{k} = 0$: $c_t = f(k_t) - (n + d + g)k_t$
- $\dot{c} = 0$: $r = \rho + \theta g$ or $k_t = k_*$, where $f'(k_*) = d + \rho + \theta g$

  Usually, $g = 0$, especially if you get the EE from the consumer max problem.

  λ never jumps at t_A (unless capital is destroyed which is rare) c always jumps at t_A (i.e., when there is a surprise). We don't know which way it jumps without more clues though.

Jump or no jump?

	$t = t_A$	$t = t_C$
Surprise change	N/A	c_t, λ_t typically jump
Pre-announced change	c_t, λ_t typically jump	c_t, λ_t typically continuous*

*Pre-announced change makes c_t, λ_t jump at the time of change if:

- Both c_t, λ_t jump if k_t jumps at time t_C for an exogenous reason (e.g. pre-announced confiscation of wealth)

- c_t jumps if parameter changes make u_c, Γ_c discontinuous (e.g. pre-announced jump in sales tax rate or a parameter change making marginal utility jump)

Algorithm to solve Ramsey Shocks

1. Transition K to k
2. Solve Planner's Problem
    1. Derive EE
    2. Check that you have two DEs (should look like this):
        1. $\frac{\dot{c_t}}{c_t} = \alpha k^{\alpha-1} - d - \rho$
        2. $\dot{k_t} = k_t^\alpha - c_t - (n+d)k_t$
3. Derive isoclines. How do they move (if at all)?
    1. $\dot{c} = 0 \Rightarrow k = (\frac{\alpha}{d+\rho})^{\frac{1}{1-\alpha}}$ (hint: make sure exponent is positive)
    2. $\dot{k} = 0 \Rightarrow c_t = k_t^\alpha - (n+d)k_t$
4. Is there a jump in c_t at t_c?
    1. Capital not destroyed, so co-state is continuous
    2. Set $\lambda_{t-0} = \lambda_{t+0}$ to see if c_t is continuous or not.
5. Draw diagram
    1. old/new curves + old/new arrows
    2. where should you be at t_{c-0} to be able the new stable arm?
    3. what is plausible behavior at $t = t_A$? Options are
        1. stay
        2. jump up
        3. jump down (can't always choose which one–sometimes either 2 or 3 is correct in drawings)
    4. Label jumps and their timing

Ramsey - Competitive Equilibrium/Household Expectations

- c_{it}: per-capita consumption (not consumption per effective worker)
- g: tech growth. We usually assume $g = 0$ - no technology growth in the consumer problem so the Euler equations of the Household and Planner coincide.
- ε_u: elasticity of Marginal Utility growth with respect to consumption growth. If ε_u rises, then consumption growth is less sensitive to changes in the interest rate. In other words, consumers prefer smooth consumption despite time-varying interest rates.
- $\Phi(K)$: an expectation/belief function that allows households to forecast future factor prices

Household Utility Maximization

$$\max_{c_{it}} \int_0^\infty e^{-\rho t} L_{it} u(c_{it}) dt \text{ s. to:}$$

1. $\dot{A_{it}} = r_t A_{it} + w_t L_t - c_{it} L_{it}$ (aka household flow budget constraint) DON'T USE $\dot{K}$ here!!
2. A_{i0}: given
3. $\lim_{T\to\infty} e^{-\bar{r}(0,T)} A_{it} \geq 0$

- Individual State Variable: A_{it}
- Aggregate state variables: can use r_t, w_t or K_t, L_t. Both will give the same answer, but we use K_t, L_t in this class.

Household's Utility Maximization Problem FOCs:

- $\frac{\partial \mathcal{H}}{\partial c_{it}} = 0 \iff$
- $\frac{\partial \mathcal{H}}{\partial A_{it}} = -\dot{\mu_{it}}$

Then eliminate μ_t, do some nasty rearranging and define $\varepsilon_u = |\frac{u''(c_{it})c_{it}}{u'(c_{it})}|$

Euler Equation for Consumption (Solution to the Ramsey Problem)

$$\frac{\dot{c_{it}}}{c_{it}} = \frac{1}{\varepsilon_u}(r_t - \rho) = -\frac{u'(c_t)}{u''(c_t)}(r_t - \rho)$$

This is identical to the Euler equation in the planner's problem if $g = 0$ and $\theta = \varepsilon_u$.

Ramsey Asset Market Clearing

$$A_t = K_t$$

Technically also have the capital leasing sector which requires

$$A_D = K_t \text{ and } L_D = L_t$$

Ramsey Capital leasing max problem Idk if this is important, so don't get sidetracked, but here it is:

$$\max_{A_D}(R_t - r_t - d)A_D$$

where A_D is the decision of the capital leasing firms.

Ramsey Goods Market Clearing

$$Y_t = C_t + I_t$$

Note: Asset Market Clearing implies goods market clearing in Ramsey

Def "Competitive (Walrasian) Equilibrium"

Competitive equilibrium is a list of quantities $\{C_t, Y_t, I_t, K_t, A_t\}$ and prices $\{w_t, r_t, R_t\}$ such that

1. $\{C_t, A_t\}$ solves the household problem given $\{w_t, r_t\}$
2. Production side equilibrium holds for all t
    1. $w_t = \frac{\partial Y_t}{\partial L_t}, R_t = \frac{\partial Y_t}{\partial K_t}, r_t = R_t - d$
3. Market clearing for all t
    1. $A_t = K_t \Rightarrow Y_t = C_t + I_t$

Def "Production Side Equilibirum"

A list of quantities $\{C_t, Y_t, K_t\}$ and prices $\{w_t, r_t, R_t\}$ satisfying, for all t:

1. Aggregate production function: $Y_t = F(K_t, L_t)$
2. Profit maximization and factor market clearing
    1. Final goods sector: $(K, L) \in argmax_{(K,L)}(F(K, L) - R_t K - w_t L)$ given (w_t, R_t)
    2. Value maximization and asset market clearing, capital leasing: $K_t \in argmax_K(\int_t^\infty e^{-\bar{r}(t,s)-d(s-t)} R_s K ds - K)$ given (r_t, R_t)
3. Free entry (automatic given profit maximization and market clearing)
    1. $\max_{(K,L)}(F(K, L) - R_t K - w_t L) = 0$

Def "Household Agrees with the Planner"

When we have (1) production side equilibrium, and (2) Asset market clearing ($A = K$), then planner and household have the same (1) utility, (2) LOM for wealth ($\dot{K}$), (3) transversality condition.

Careful: if we don't have asset market clearing and/or production side equilibrium, HH would have different resources and the stable arms would not coincide.

Def "Rational Expectations Equilibrium (REE)"

Refinement of WE - we add an expectation/belief function $\Phi(K) = c_t, C_t = \Phi(K_t)L_t$ that allows households to forecast future factor prices

Definition of equilibrium: REE is a list of quantities, $\{C_t, Y_t, I_t, K_t, A_t\}$, prices $\{w_t, r_t, R_t\}$ and $\Phi(K)$ satisfying

1. Utility maximization (*different from WE*) The consumption decision rule $c_i(a, K_t; \Phi(K_t))$ solves the household problem given the information households have at date t and their beliefs about behavior of others, $c_t = \Phi(K_t)$

Details: Households form factor price forecasts (function-to-function mappings) based of date-t state K_t and their beliefs Φ:

$$\{w^e_{t+\tau}\} = \mathcal{W}(K_t; \Phi(K_t)), \qquad \{r^e_{t+\tau}\} = \mathcal{R}(K_t; \Phi(K_t)), \text{all } \tau > 0.$$

Utility maximization problem:

$$c_i(a, K_t; \Phi) \in \underset{\{c_{it}\}}{\operatorname{argmax}} \int_0^\infty e^{-\rho t} L_{it} u(c_{it}) dt$$

$$\int_t^\infty \exp[-\bar{r}^e(t,s)] \cdot c_{is} \cdot L_{is} ds \leq a \cdot L_{it} + \int_t^\infty \exp[-\bar{r}^e(t,s)] \cdot w^e_s \cdot L_{is} ds$$

Definition of equilibrium – continued

2. Production-side equilibrium (no change)

3. Rational expectations (consistency) <u>conditions</u> <u>on Φ</u> (new)

Households' own consumption decision (c_i) based on *equilibrium* beliefs is consistent with their beliefs about others' consumption (Φ) for any initial condition $K_0 > 0$ and all $t \geq 0$:

$$c_i\left(\frac{K_t}{L_t}, K_t; \Phi(K_t)\right) = \Phi(K_t)$$

where K_t solves

$$\dot{K}_t = F(K_t, L_t) - \Phi(K_t)L_t - dK_t \text{ given } K_0 > 0, L_t = L_0 e^{nt}.$$

4. Asset market clearing (no change), follows from RE condition 3

FINISH MEE

Def "Representative Household flow budget constraint"

$$\dot{A}_t = r_t A_t + w_t L_t - C_t$$

Government and fiscal policy in the Ramsey model

Endogenous Growth

- A: (stock?) patent number (index)
- $P_{A,t}$: price of idea
- M (?) market value of patents
- $\dot{M}$: Capital gain
- φ: Knowledge externality. Stepping on toes or synergy.
- λ: Labor externality...
- θ: ????????????????????????????
- flow = quantity measured as bricks per quarter
- stock = total number of bricks

Misc

On the balanced growth path, all components of expenditure have to grow at the same rate as Y

Def "Euler equation"

An Euler equation is something that relates choice in one period to choice in another period.

Remark "discrete to continuous, stocks vs flows"

REMEMBER HOW TO CONVERT - one of them gets multiplied by delta t

In steady state $\dot{K} \neq 0$!!!! ONLY RATIOS ARE UNCHANGING IN STEADY STATES!!

Prereqs

Def "Annual growth rate"

Def "Annualized growth rate"

Def "Instantaneous growth rate"

Fact "Arithmetic with growth rates"

- $g_{XY} = g_X + g_Y$
- $g_{X/Y} = g_X - g_Y$
- $g_{X^\alpha} = \alpha g_X$